引题001

一. $\mu=0.5$

a). 底不动. U_0.?

$U_0^2 = 2g\mu \cdot 2l$ ×

支持力等效作用点 重心右 Δ.

角动量定理.

① 以H为支 不转 $\Rightarrow \Sigma M=0$ $\Delta=0$?

地系: $\frac{\Delta \frac{ml}{2}v}{\Delta t} = m\frac{l}{2}a = N\cdot\Delta$

② 质心系中 以H为支 $\Sigma M=0$ 有摩擦力 力矩

$\Rightarrow ma\cdot\frac{l}{2} = N\cdot\Delta$

牛二 $ma = mg\mu$

$\Rightarrow \Delta = \frac{ma\frac{l}{2}}{mg} = \frac{l}{4}$.

当支持力作用点到边上 "掉" → 翻

匀加速: $U_0^2 = 2\cdot g\mu\cdot(2l-\Delta)$.

$U_0 = \sqrt{1.75gl}$.

②. $\mu = 1-\frac{h}{4l}$. $h\in(0,4l)$.

$\begin{cases}\mu_1 = 1-\frac{x+\frac{l}{2}}{4l} = -\frac{x}{4l}+\frac{7}{8} \\ \mu_2 = 1-\frac{x-\frac{l}{2}}{4l} = -\frac{x}{4l}+\frac{9}{8}\end{cases}$

$N = N_1+N_2 = mg$.

牛二 $ma = -N_1\mu_1 - N_2\mu_2$

从C为系 以C为支: $\Sigma M=0$. (不动力)

$N_2\frac{l}{2} - N_1\frac{l}{2} = (N_1\mu_1+N_2\mu_2)\frac{l}{2}$.

$\Rightarrow \begin{cases} N_2-N_1 = N_1\cdot(-\frac{x}{4l}+\frac{7}{8}) + N_2(-\frac{x}{4l}+\frac{9}{8}) \\ N_1+N_2 = mg. \end{cases}$

$\begin{cases} N_2\cdot(-\frac{1}{8}+\frac{x}{4l}) + N_1(-\frac{15}{8}+\frac{x}{4l}) = 0 \\ N_1+N_2 = mg \end{cases}$

$N_1 = -\frac{-\frac{1}{8}+\frac{x}{4l}}{\frac{x}{2l}-2}mg$; $N_2 = \frac{-\frac{15}{8}+\frac{x}{4l}}{\frac{x}{2l}-2}mg$ $2-\frac{x}{2l}>0$

$\Rightarrow ma = -m\ddot{x} = -N_1\mu_1 - N_2\mu_2$

$-\frac{m\ddot{x}}{mg} = \frac{\frac{x}{4l}-\frac{1}{8}}{2-\frac{x}{2l}}\cdot(-\frac{x}{4l}+\frac{7}{8}) - \frac{\frac{15}{8}-\frac{x}{4l}}{2-\frac{x}{2l}}\cdot(-\frac{x}{4l}+\frac{9}{8})$

$7\ddot{x} = 8g - 2x\frac{g}{l}$ ⇒ 简振. $\omega^2 = \frac{2g}{7l}$.

$t=0$. $x=2l$. $\dot{x}=-U_0$

$x = 4l + A\cos(\omega t+\varphi)$.

$\begin{cases} 2l = 4l + A\cos\varphi \\ -U_0 = -\omega A\sin\varphi \end{cases}$

$\Rightarrow A = \sqrt{(2l)^2+(\frac{U_0}{\omega})^2} = \frac{7}{2}l$. $\Rightarrow U_{0\max} = \sqrt{\frac{33}{14}gl}$

$x-4l = A\cos(\omega t+\varphi)$

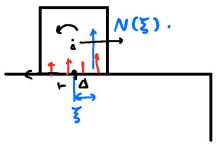
$-\dot{x}_0$

$U_0\cos\theta$ θ $2l$ $x-4l$. $x_{\min}=\frac{1}{2}$.

$\mu = 1-\frac{h}{4l}$; (只有两点.)

设分布 $dN = N(\xi)\,d\xi$.

↳ 单位长度支持力.



① $\int_{-l/2}^{\frac{l}{2}} N(\xi)\,d\xi$ ← $\Sigma M=0$. 以C为支 以C为系.

$= \int_{-l/2}^{l/2}(1-\frac{x-\xi}{4l})N(\xi)\,d\xi\cdot\frac{1}{2}$.

↳ 算 Δ.

临界 $\theta = \arccos\frac{2l}{\frac{7}{2}l}$

$t = \frac{\theta}{\omega} = \sqrt{\frac{7l}{2g}}\cdot\arccos\frac{4}{7}$

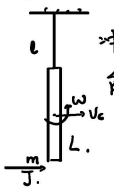
② $mg = \int_{-l/2}^{l/2} N(\xi)\,d\xi$

③ $-m\ddot{x} = \int_{-l/2}^{l/2}(1-\frac{x-\xi}{4l})N(\xi)\,d\xi$

mg $\bar{t}$ $mg\Delta$

$A = \int_{\mu_2}^{\mu_1} N(\xi)\,d\xi$; $B = \int_{-l/2}^{l/2} N(\xi)\xi\,d\xi$. ↑ 代入

二. ① 打末端. 求 v_c. ω.

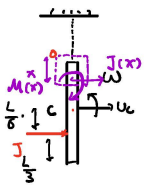


冲量: $mv_c = J$.

角动量: 以O为支 (以C为支)

$J \cdot L = mv_c \frac{L}{2} + \frac{1}{12}mL^2 \cdot \omega$ ($J\frac{L}{2} = \frac{1}{12}mL^2\omega$).

$$\Rightarrow \begin{cases} v_c = \frac{J}{m} \\ \omega = \frac{6J}{mL} \end{cases}$$



② 打 1/3 处.

以C为支.

$$\begin{cases} \frac{L}{6} \cdot J = \frac{1}{12}mL^2\omega \\ J = mv_c \end{cases}$$

$\Rightarrow v_c = \frac{J}{m}$. $\omega = \frac{2J}{mL}$. ($v_0 = v_c - \omega\frac{L}{2} = 0$)



取从上向下 x 高为对象. 写冲量定理.

$J(x) = m \cdot \frac{x}{L} \cdot \omega \cdot \frac{x}{2}$ (纯冲量 = 0)

$\Rightarrow J(x) = m \cdot \frac{x^2}{2L} \cdot \frac{2J}{mL} = \frac{x^2}{L^2}J$.

角动量: 以O为支. x.

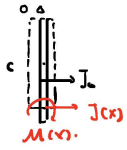
$+M(x) + \omega \cdot \frac{1}{3} \cdot \frac{x}{L}m \cdot x^2 = J(x) \cdot x$. 不等?

$M(x) = -\frac{2J}{mL} \cdot \frac{m}{3L}x^3 + \frac{x^2}{L^2}J \cdot x$

$= +\frac{J}{3L^2}x^3$ $x \in (0, \frac{2L}{3})$.

$\{M(x)\}_{max} = \frac{J}{3L^3} \cdot \frac{8L^3}{27} = \frac{8}{81}JL$.

取从上向下 x 高为对象. 写冲量定理. $x > \frac{2L}{3}$



冲量:

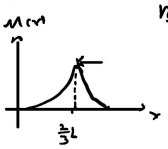
$J(x) + J_0 = \frac{mx}{L} \cdot \omega \cdot \frac{x}{2}$.

冲量矩:

$\frac{mx}{L} \cdot \frac{1}{3}x^2 = J_0\frac{2L}{3} + J(x) \cdot x - M(x)$. ← 单调.

$\Rightarrow \{M(x)\}_{max} = \frac{8}{81}JL$.

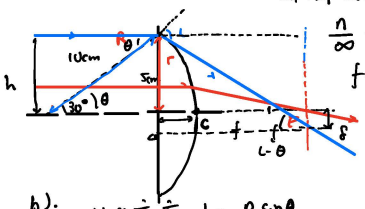
$x = \frac{2}{3}L$.



$\Rightarrow J_{max} = \frac{M_0 81}{L\ 8}$.

三. $n=1.58$, $R=10cm$, $r=5cm$

a). 单球面成像



$\frac{n}{\infty}+\frac{1}{f}=\frac{n-1}{R}$

$f=\frac{R}{n-1}=17.24cm$

L 近轴

b). 从θ高度 $h=R\sin\theta$

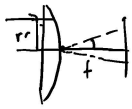
$n\sin\theta=\sin L$

$\Rightarrow \delta=[f+(10cm-5\sqrt{3}cm)]\cdot\tan(L-\theta)-R\sin\theta$.

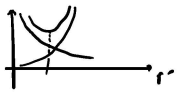
(可证. $\frac{d\delta}{d\theta}\geq 0$ 单调.) → 打表. $\delta-\theta$ 发现.

$\delta(\theta=30°)_{max}=2.6cm$.

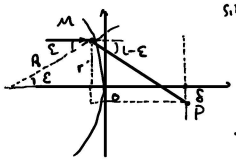
c). $\delta'=f\cdot\frac{1.22\lambda}{2\cdot r'}$ ~ δ. 小角.



($\delta+\delta'$ 打表发现)



以r'为变量. 小角.



$\sin\varepsilon=\frac{r'}{R}$

$M\cdot(-R+R\cos\varepsilon, R\sin\varepsilon)$.

$\sin L=n\sin\varepsilon$.

MP 斜率 $k=-\tan(L-\varepsilon)$.

$\delta(\varepsilon)=-\delta(-\varepsilon)$

$\delta=[\frac{R}{n-1}+R(1-\cos\varepsilon)]\tan(L-\varepsilon)-R\sin\varepsilon$

$O(1)$ $O(2)$ $O(3)$ $O(3)$ $O(3)$

$O(1)\to 0$ $O(2)=0$. 奇函数. $O(3)$. 打表得.

$\delta=(\frac{R}{n-1}+R\frac{1}{2}\varepsilon^2)\cdot\frac{\tan L-\tan\varepsilon}{1+\tan L\tan\varepsilon}-R(\varepsilon-\frac{1}{6}\varepsilon^3)$

$O(1)+O(3)$

$$\frac{\frac{n\sin\varepsilon}{\sqrt{1-n^2\sin^2\varepsilon}}-\tan\varepsilon}{1+\frac{n\sin\varepsilon}{\sqrt{1-n^2\sin^2\varepsilon}}\tan\varepsilon}=\frac{n\sin\varepsilon-\sqrt{1-n^2\sin^2\varepsilon}\cdot\tan\varepsilon}{\sqrt{1-n^2\sin^2\varepsilon}-n\frac{\sin^2\varepsilon}{\cos\varepsilon}}$$

$$=\sin\varepsilon\frac{n-(1-\frac{1}{2}n^2\varepsilon^2)(1+\frac{1}{2}\varepsilon^2)}{1-\frac{1}{2}n^2\varepsilon^2-n\cdot\varepsilon^2}$$

$$=\varepsilon(1-\frac{1}{6}\varepsilon^2)\cdot\frac{n-1+\frac{1}{2}n^2\varepsilon^2-\frac{1}{2}\varepsilon^2}{1-(\frac{1}{2}n^2+1)\varepsilon^2}$$

$$=(n-1)\varepsilon\cdot(1-\frac{1}{6}\varepsilon^2+(\frac{1}{2}n^2+1)\varepsilon^2+\frac{n+1}{2}\varepsilon^2)$$

$$\delta=(\frac{R}{n-1}+R\frac{1}{2}\varepsilon^2)\cdot(n-1)\varepsilon\cdot(1+(-\frac{1}{6}+\frac{1}{2}+\frac{3}{2}n+\frac{n^2}{2})\varepsilon^2)-R(\varepsilon-\frac{1}{6}\varepsilon^3)$$

$$=\varepsilon^3\cdot[\frac{R}{6}+R\frac{1}{2}(n-1)+\frac{R}{n-1}(-\frac{1}{3}+\frac{3}{2}n+\frac{n^2}{2})]$$

$$=\alpha\cdot R\cdot\varepsilon^3.$$

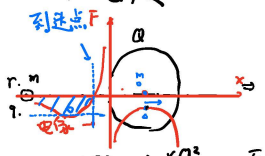
$\delta'=\frac{R}{n-1}\cdot\frac{1.22\lambda}{2\cdot R\varepsilon}$; $\delta+\delta'=\alpha R\varepsilon^3+\frac{0.61\lambda}{n-1}\cdot\frac{1}{\varepsilon}$.

均值: 当 $\alpha R\varepsilon^3=\frac{1}{3}\cdot\frac{0.61\lambda}{n-1}\frac{1}{\varepsilon}$ 时. 最小.

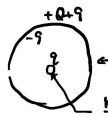
$\varepsilon=(\frac{1}{\alpha R}\cdot\frac{1}{3}\frac{0.61\lambda}{n-1})^{\frac{1}{4}}\Rightarrow\varepsilon=0.035$. ✓

$(\delta+\delta')_{min}=2.2\times10^{-5}m$. ✓

四. Q, R M. r<<R.



$E_p = \frac{1}{2}\frac{kq^2}{r} + \frac{1}{2}\frac{kQ^2}{R}$.



$\frac{1}{2}\cdot$ 27:U:

$\frac{k(Q+q)}{R}$ ← 叠加

$\frac{kQ}{R} + \frac{kq}{r}$ ←

$E_p = \frac{1}{2}q\cdot\left(\frac{kQ}{R}+\frac{kq}{r}\right) + \frac{1}{2}Q\cdot\frac{k(Q+q)}{R}$

$= \frac{1}{2}\frac{kq^2}{r} + \boxed{\frac{kQq}{R}} + \frac{1}{2}\frac{kQ^2}{R}$.

$\Delta E_p = \frac{kQq}{R}$.

$\frac{1}{2}mv_0^2 = \frac{kQq}{R} + \frac{1}{2}mv_m^2 + \frac{1}{2}Mv_M^2$

$v_m = v_c + v_{mr}$; $v_M = v_c + v_{Mr}$ } 代入

$v_{mr} = -\frac{M}{m+M}v_r$; $v_M = \frac{m}{m+M}v_r$.

$\frac{1}{2}\frac{mM}{m+M}v_r^2 = \frac{1}{2}\frac{mMv_0^2}{m+M} - \frac{kQq}{R}$.

$v_r = \pm\sqrt{v_0^2 - \frac{2kQq(m+M)}{RmM}}$

$v_m = \frac{mv_0}{m+M} \mp \frac{M}{m+M}\sqrt{v_0^2 - \frac{2kQq(m+M)}{RmM}}$ +,-,± ✓

$v_m = v_c + v_r$. 取正根.

(1). 电荷到表面:



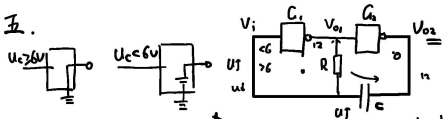
$E_p = \frac{1}{2}\frac{k(Q+q)^2}{R}$.

$\Delta E = -\frac{1}{2}\frac{k(Q+q)^2}{R} + \frac{kQ^2}{2R} + \frac{kq^2}{2r}$.

$= -\frac{kQq}{R} + \frac{kq^2}{2r} - \frac{kq^2}{2R}$.

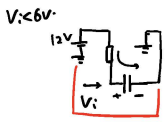
(2) $= \frac{1}{2}\frac{mM}{m+M}v_0^2 + \frac{kq^2}{2r} - \frac{kq^2}{2R} - \frac{kQq}{R}$

五.



$U_c(t) = U_0(1 - e^{-\frac{t}{RC}})$. 无稳态. 动态过程.

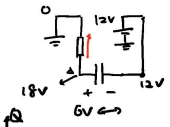
① 设初态. $Q_c = 0$ $V_i = 0$.



充电. $U_c = 12V \cdot (1 - e^{-\frac{t}{RC}})$.

$\rightarrow t = ?$ $V_i = 6V$. $Q_c = C \cdot 6V$.

② $V_i = 6V$. 翻转. 电量不突变

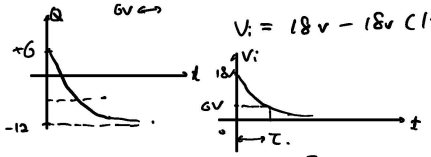


$V_i = 6V \rightarrow 18V$.

末态 $Q_c = -12V \cdot C$. 初 $Q_c = 6V \cdot C$

$Q_c = 6V \cdot C - 18V \cdot (1 - e^{-\frac{t}{RC}})$.

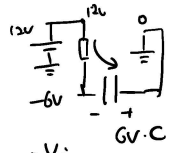
$V_i = 18V - 18V(1 - e^{-\frac{t}{RC}})$.



到 $V_i = 6V$; $e^{-\frac{t}{RC}} = \frac{1}{3}$ $\tau = RC\ln 3$.

此后 $V_i < 6V$.

③ 再翻 $Q_c = -6V \cdot C$.



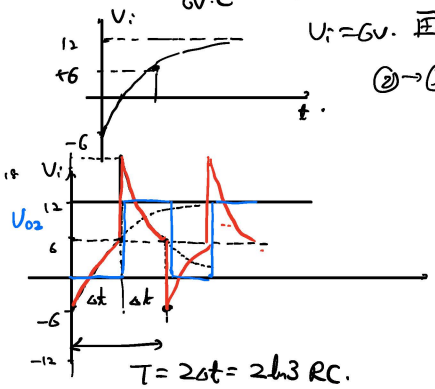
$V_i = 6V \rightarrow -6V$.

$Q_c = -6V + 18V \cdot (1 - e^{-\frac{t}{RC}})$.

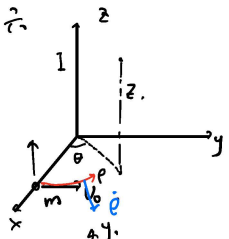
$V_i = -6V + 18V(1 - e^{-\frac{t}{RC}})$

$V_i = 6V$. 再翻 $\rightarrow$ ②

② $\rightarrow$ ③ $\rightarrow$ ② $\rightarrow$ ...



$T = 2\Delta t = 2\ln 3\, RC$.

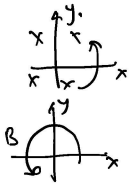


$I \rightarrow B = \frac{\mu_0 I}{2\pi\rho}$.

① E_k ✓. $B \perp \hat{\theta}$

$\frac{dL_z}{dt} = F_\theta \cdot \rho = 0$

L_z 守恒.



② $m\frac{dv_z}{dt} = q\dot{\rho} B_\theta$

$= q\frac{d\rho}{dt} \cdot \frac{\mu_0 I}{2\pi\rho}$

$\Rightarrow m\,dv_z = q\frac{\mu_0 I}{2\pi}\frac{d\rho}{\rho}$.

$\Rightarrow d\left[mv_z - q\frac{I\mu_0}{2\pi}\ln\rho\right] = 0$

$mv_z - q\frac{I}{2\pi}\ln\rho = 0 - q\frac{I}{2\pi}\ln r_0$

$mv_z = q\frac{I}{2\pi}\ln\frac{\rho}{r_0}$

③ 最远 $\dot{\rho}=0$. $\frac{r_m}{r_0} = \lambda$.

$\Delta L_z = 0$. $m v_t r_m = m v_0 r_0$.

$v_t = v_0 \cdot \frac{1}{\lambda}$.

$\Delta E = 0$.

$\frac{1}{2}m \cdot v_0^2 = \frac{1}{2}mv_z^2 + \frac{1}{2}m\dot{\rho}^2 + \frac{1}{2}mv_t^2$

$= \frac{1}{2}\cdot\frac{\left(q\frac{I\mu_0}{2\pi}\ln\lambda\right)^2}{m} + \frac{1}{2}m v_0^2\frac{1}{\lambda^2}$.

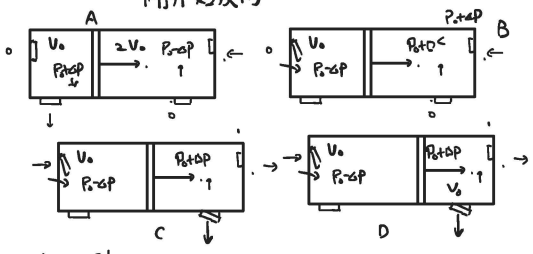
$\Rightarrow 1 = \left(\frac{q}{m}\right)^2\left(\frac{\mu_0 I}{2\pi}\right)^2\ln^2\lambda + \frac{1}{\lambda^2}$

$1 = \frac{1}{4}\ln^2\lambda + \frac{1}{\lambda^2}$; $\lambda_1 = 1$. $\lambda_2 = 7.25$

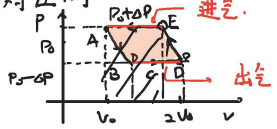
$r_m = 7.25\, r_0$.

七.

刚开始反向



对左侧



① 一个循环.

左侧: $\Delta n = \frac{1}{RT}(P_D \cdot V_D - P_B \cdot V_B)$

$= \frac{1}{RT} \cdot (P_0 V_0 - P_A V_A)$

$= \frac{1}{RT} \cdot \{(P_0-\Delta P)\cdot 2V_0 - (P_0+\Delta P)\cdot V_0\}$

$= \frac{P_0 V_0}{RT} \cdot (1 - \frac{3\Delta P}{P_0})$.

$(P_0-\Delta P)\cdot 2 > P_0+\Delta P$

$\Delta P < \frac{P_0}{3}$.

总 $\Delta n_{总} = 2\Delta n = \frac{2P_0V_0}{RT}(1-\frac{3\Delta P}{P_0})$.

② 左对气体做功 $\times 2$ = 外力对活塞 $\frac{nRT}{U}dU$

$W = 2\cdot[(P_0+\Delta P)\cdot[V_E - V_0] + \int_D^E p\,dV$

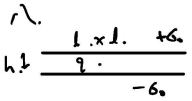
$- (P_0-\Delta P)\cdot(2V_0 - V_B) + \int_A^B p\,dV]$

$= 2\cdot[2V_0(P_0-\Delta P) + 2V_0(P_0-\Delta P)\ln\frac{P_0+\Delta P}{P_0-\Delta P}$

$- V_0(P_0+\Delta P) + V_0(P_0+\Delta P)\ln\frac{P_0+\Delta P}{P_0-\Delta P}]$.

$= 2V_0(P_0-3\Delta P)\cdot\ln\frac{P_0+\Delta P}{P_0-\Delta P}$

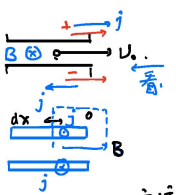
八.



① $F_q = \frac{\sigma_0}{\varepsilon_0} \cdot q$.

② $\sigma_0 \to \sigma_0 \cdot \frac{1}{\sqrt{1-u_0^2/c^2}}$

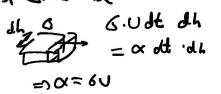
$E = \frac{\sigma_0}{\varepsilon_0} \frac{1}{\sqrt{1-u_0^2/c^2}}$.



③ $dI = \alpha \cdot dx$ ← 线电流密度

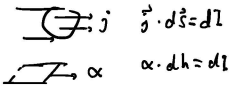
$\alpha = \sigma \cdot u$

$= \frac{\sigma_0}{\sqrt{1-u_0^2/c^2}} u_0$



$\sigma \cdot u\,dt\,dh = \alpha\,dt\,dh$

$\Rightarrow \alpha = \sigma u$



$\vec{j} \cdot d\vec{S} = dI$

$\alpha \cdot dh = dI$

安培环定理

$B \cdot = \mu_0 \cdot \frac{\sigma_0}{\sqrt{1-u_0^2/c^2}} u_0$

$F' = \frac{\sigma_0}{\varepsilon_0} \frac{1}{\sqrt{1-u_0^2/c^2}} - u_0 \cdot \frac{\mu_0 \sigma_0}{\sqrt{1-u_0^2/c^2}} u_0$ $\quad \mu_0\varepsilon_0 = \frac{1}{c^2}$ 代入

$= \frac{\sigma_0}{\varepsilon_0} \frac{1}{\sqrt{1-\frac{u_0^2}{c^2}}} (1 - u_0^2/c^2)$

$= \frac{\sigma_0}{\varepsilon_0} \sqrt{1-u_0^2/c^2}$

$dP_2 = F_2\,dt$ $\quad dP_2' = F_2'\,dt'$ ✓ $\quad dt = \sqrt{1-u_0^2/c^2} \cdot dt'$ 代入

↑ 原 $\quad$ 新 $\quad \Rightarrow dP_2 = dP_2'$ 自洽.

③' → 1). $F_{板} = \frac{1}{2} \varepsilon_0 E^2 \cdot S$

$= \frac{1}{2} \frac{\sigma_0^2}{\varepsilon_0} \cdot l^2$

2). $F_{板}' = F_{板} \sqrt{1-u_0^2/c^2}$ ✓ $(F_{板}'\,dt' = F_{板}\,dt)$

$\frac{F_{板}'}{S'} = \frac{1}{2} \sigma \cdot E' - \frac{1}{2} B' \cdot \alpha'$ 代入. $\quad \gamma = \frac{1}{\sqrt{1-u_0^2/c^2}}$

$= \frac{1}{2} \sigma_0 \gamma \cdot \frac{\sigma_0}{\varepsilon_0} \cdot \gamma - \frac{1}{2} \cdot \frac{u_0}{c^2} \cdot \frac{\sigma_0 \gamma}{\varepsilon_0} \cdot \sigma_0 \cdot u_0$

$= \frac{1}{2} \sigma_0^2 / \varepsilon_0 \cdot (\gamma^2 - \gamma^2 \frac{u_0^2}{c^2})$

$= \frac{1}{2} \frac{\sigma_0^2}{\varepsilon_0} \cdot 1$ $\quad S' = \frac{1}{\gamma} S$.

$F_{板}' = \frac{1}{2} \frac{\sigma_0^2}{\varepsilon_0} \cdot l^2 \cdot \sqrt{1-u_0^2/c^2}$